

Tools To Tackle Your DIGITAL And ELECTRONIC DESIGN CHALLENGES

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Digital Logic Design

This is a software tool for designing and simulating digital circuits. It provides digital parts ranging from simple gates to arithmetic logic units and state machines. Using this software, circuits can easily be converted into reusable modules, which may be used to build more complex circuits like CPUs. This software requires Java 1.7 or above.

McCAD tools

McCAD tools for electronic designs include three main software components, namely, EDS-Plus Bundle, SimPlus Bundle and 3SPICE. EDS-Plus Bundle is a fully-operational electronic design automation (EDA) system that produces manufacturable outputs. SimPlus Bundle is a fully-operational EDA system that allows you to create and simulate circuits. 3SPICE is a fully-functional circuit simulator based on industry-standard Berkeley SPICE3.

Circuit Sandbox (version 1.0)

Circuit Sandbox allows you to visually build Boolean logic circuits and then simulate their operation. You can create custom components from user-designed circuits. The software is written in Java for cross-platform functionality.

ZenitPCB (version 1.8.1)

ZenitPCB suite is for making printed circuit boards (PCBs). It comprises two main components: ZenitCapture and ZenitPCB. ZenitCapture schematic is an easy-to-use tool for creating schematic designs swiftly. ZenitPCB is a tool to create professional-grade PCB layouts. It is completely free to use, limited to 800 pins.

ZenitPCB GerberView can be used to view the created Gerber file.

Some popular resources

WinSCP (version 5.7.6). WinSCP is an open source, free SFTP client, FTP client, WebDAV client and SCP client for Windows. Its main function is file transfer between a local and a remote computer. The latest version improves interoperability with several SFTP and FTP servers and fixes several bugs.

Angry IP Scanner (version 3.4). This is a very fast IP address and port scanner. It can scan IP addresses in any range as well as any ports. It does not require any installation, and can be freely copied and used anywhere.

Password Safe (version 3.37.1). Password Safe is a simple and secure password-management application available under open source approved licence.

Eraser (version 6.0.10.2620). Eraser is an advanced security tool for Windows that allows you to completely remove sensitive data from your hard drive by overwriting it several times with carefully-selected patterns.

KeePass (version 2.30). A free, open source, lightweight and easy-to-use password manager. It helps manage passwords in a secure way. You can put all passwords in one database, which is locked with one master key or key file. You only have to remember one master password or select the key file to unlock the database.

EFY anniversary special

DipTrace (version 2.4.0.2) six-month trial included in the DVD. DipTrace, the schematic and PCB design software, can now be used for a period of 180 days from the date of installation. Users can access all features and libraries. The 3D library package is also included as an additional file for real-time 3D preview and export.

Note. A freeware version of DipTrace (version 2.4) has also been included in the DVD for Mac OS X users. Although the designs are limited to 300 pins and two signal layers.



Fig. 1: Interface of Digital Logic Design tool

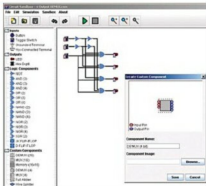


Fig. 2: Creating 4-output demultiplexer component with Circuit Sandbox

Osmond PCB (version 1.1.30)

Osmond PCB offers great power and flexibility with artificial limits and restrictions being banished. Particularly designed for Macintosh computers, it allows you to design boards of any size and shape and

with as many layers as you want. With a spatial resolution of 10 nanometers, Osmond gives the precision you need. You can place parts anywhere

on the board with any orientation. You can also run traces of any width along any path and at any angle. Once your design is complete, Osmond can produce standard Gerber (RS-274X) files and Excellon drill files that fabricators use to make boards.

Elipse SCADA (version 2.29 b151)

By collecting information from any type of device, Elipse SCADA allows you to monitor and accurately control plant floor processes, machinery and resources, managing production in a quick and efficient way. Real-time data is presented graphically, allowing information to be handled in several ways such as historical queries, reports generation and remote connection. **EFY**

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